

Exercise 2: E-commerce Platform Search Function

```
import java.util.Arrays;

import java.util.Comparator;

public class Ecommerce {

    static class Product {

        int productId;

        String productName;

        String category;

        public Product(int productId, String productName, String category) {

            this.productId = productId;

            this.productName = productName;

            this.category = category;

        }

        @Override

        public String toString() {

            return productId + " - " + productName + " (" + category + ")";

        }

    }

    public static Product linearSearch(Product[] products, int targetId) {

        for (int i = 0; i < products.length; i++) {

            if (products[i].productId == targetId) {

                return products[i];

            }

        }

        return null;

    }

}
```

```

public static Product binarySearch(Product[] products, int targetId) {
    int left = 0, right = products.length - 1;

    while (left <= right) {
        int mid = (left + right) / 2;

        if (products[mid].productId == targetId) {
            return products[mid];
        } else if (products[mid].productId < targetId) {
            left = mid + 1;
        } else {
            right = mid - 1;
        }
    }
    return null;
}

```

```

public static void main(String[] args) {

```

```

    Product[] products = {
        new Product(105, "Shoes", "Fashion"),
        new Product(101, "Laptop", "Electronics"),
        new Product(110, "Watch", "Accessories"),
        new Product(102, "Phone", "Electronics"),
        new Product(108, "Bag", "Fashion")
    };

```

```

    int targetId = 102;

```

```

    Product result1 = linearSearch(products, targetId);
    System.out.println("Linear Search Result: " + result1);

```

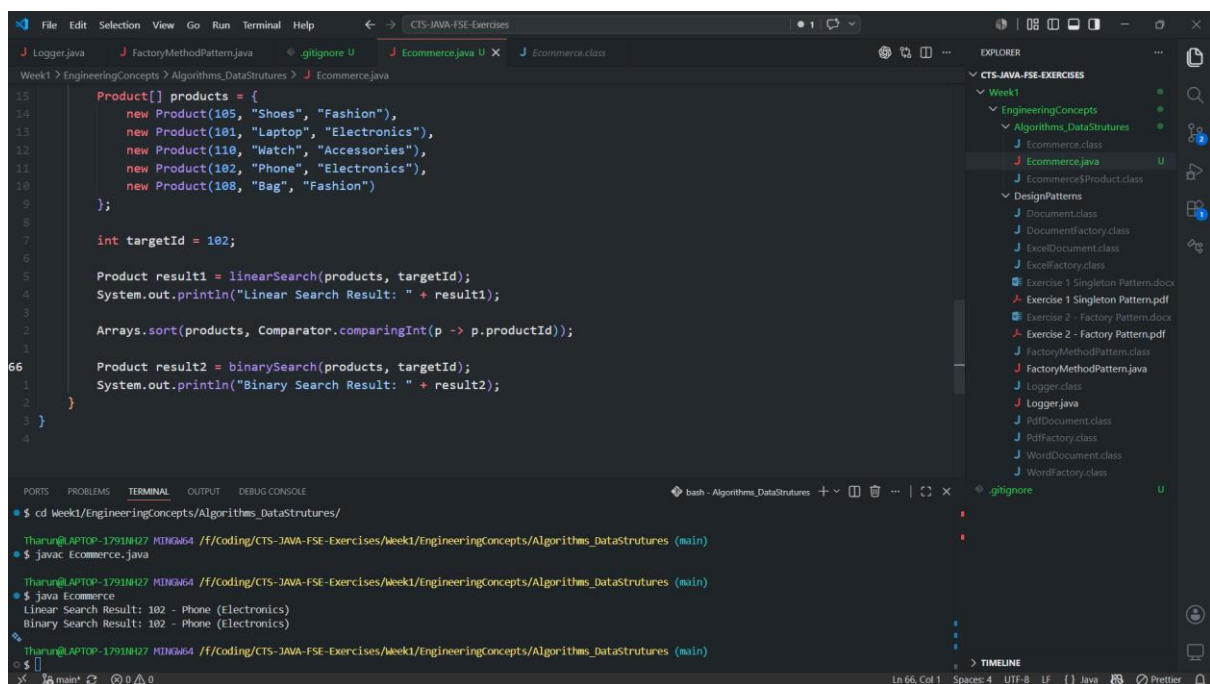
```
Arrays.sort(products, Comparator.comparingInt(p -> p.productId));
```

```
Product result2 = binarySearch(products, targetId);
```

```
System.out.println("Binary Search Result: " + result2);
```

```
}
```

```
}
```



The screenshot shows an IDE with a Java file named `Ecommerce.java` and a terminal window. The code in `Ecommerce.java` defines a `Product` array, sorts it by `productId`, and performs a linear search followed by a binary search for a target ID of 102. The terminal output shows the results of these searches.

```
15 Product[] products = {
16     new Product(105, "Shoes", "Fashion"),
17     new Product(101, "Laptop", "Electronics"),
18     new Product(110, "Watch", "Accessories"),
19     new Product(102, "Phone", "Electronics"),
20     new Product(108, "Bag", "Fashion")
21 };
22
23 int targetId = 102;
24
25 Product result1 = linearSearch(products, targetId);
26 System.out.println("Linear Search Result: " + result1);
27
28 Arrays.sort(products, Comparator.comparingInt(p -> p.productId));
29
30 Product result2 = binarySearch(products, targetId);
31 System.out.println("Binary Search Result: " + result2);
32 }
33 }
```

```
Tharun@LAPTOP-1791M427 MINGW64 /f/coding/CTS-JAVA-FSE-Exercises/Week1/EngineeringConcepts/Algorithms_DataStructures (main)
$ cd Week1/EngineeringConcepts/Algorithms_DataStructures/
$ javac Ecommerce.java
$ java Ecommerce
Linear Search Result: 102 - Phone (Electronics)
Binary Search Result: 102 - Phone (Electronics)
```